

LegalAid Call Abandonment Final Presentation

Simple EDA Team 1

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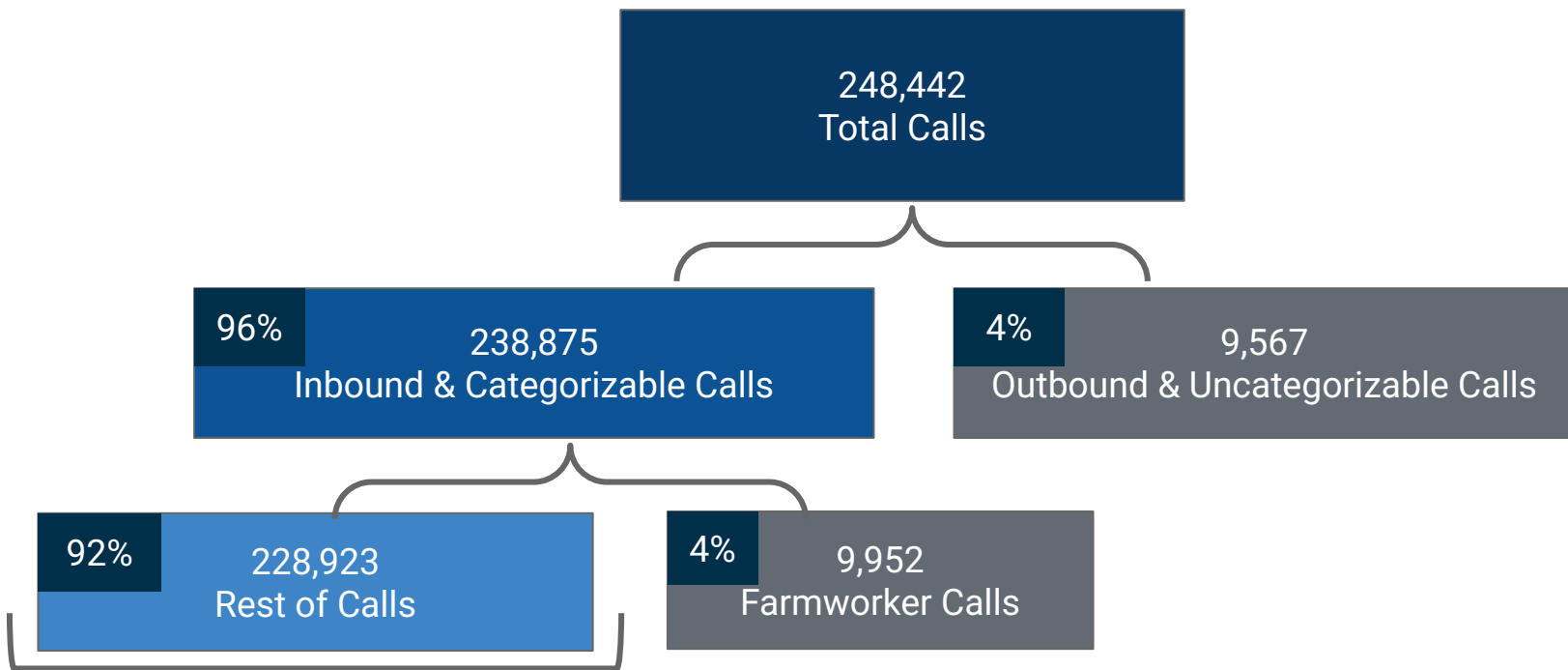
Introduction

Objective: Find overall proportion of abandoned calls, as well as proportion of abandonment by menu

Process:

- 1) Tag every call to a specific *menu* and *last activity* based on where they ended
- 2) Tag every subgroup of menu and last activity as either *abandoned* or *not abandoned*
- 3) Use these results to calculate the % of abandoned calls, both in total as well as per menu

Total Analysis Figures



Abandonment Calculations

Abandonment Calculations

Menu Name	Total Calls	% of All Calls	Abandoned Calls	Abandonment Rate
Benefits	7,210	3.15%	429	5.95%
Employment	4,564	1.99%	289	6.33%
Family	22,753	9.94%	2,242	9.85%
HIV	1,603	0.70%	767	47.85%
Housing	16,492	7.20%	3,837	23.27%
Immigration	3,516	1.54%	696	19.80%
Legal	27,715	12.11%	13,793	49.77%
Main	119,927	52.39%	40,657	33.90%
Other Legal	12,204	5.33%	6,830	55.97%
Pre-Legal Seniors	12,939	5.65%	4,425	34.20%
Totals	228,923		73,965	32.31%

Preprocessing Method – Python

- 1) New dataframe - one row per call
- 2) Per call - I found both the last menu each call passed through (of the 11 relevant menus) and I found the last 'Activity Name'
- 3) Aggregate, grouping by last menu and last activity name, finding counts for each of these subgroups
- 4) I copied my results to Google Sheets to compare with Meera's independent numbers using her method in R

Next: Each unique combination of last menu & last activity can be used to tag a subgroup of calls as abandoned or not in a MECE (mutually exclusive & collectively exhaustive) manner, powering our analysis

Menu	Last Activity Name	Count	Abandoned?
Farmworker Main Number Telephony EP	FarmworkerMainMenu	9640	
Farmworker Main Number Telephony EP	LanguageSelectionMenu	156	
Farmworker Main Number Telephony EP	MigrantVoicemailTransfer	67	
Farmworker Main Number Telephony EP	DisconnectContact1	44	
Farmworker Main Number Telephony EP	ClosedMenu	29	
Farmworker Main Number Telephony EP	ClinicVoicemailTransfer	7	
Farmworker Main Number Telephony EP	StaffDirectorySpanishTransfer	5	
Farmworker Main Number Telephony EP	StaffDirectoryEnglishTransfer	3	
Farmworker Main Number Telephony EP	AddressFaxHoursMenu	1	
Legal Benefits Menu Telephony EP	ClosedQueueMenu	4674	
Legal Benefits Menu Telephony EP	ClinicVoicemailTransfer	818	

```
df_final['Menu'].value_counts()
```

```
Main Number Telephony EP      120093
Legal Menu Telephony EP        27715
Legal Family Menu Telephony EP  22753
Legal Housing Menu Telephony EP 16492
Pre-Legal Menu Seniors Menu Telephony EP 12944
Other Legal Menu Telephony EP  12204
Farmworker Main Number Telephony EP 9953
Legal Benefits Menu Telephony EP  7210
Legal Employment Menu Telephony EP 4564
Legal Immigration Menu Telephony EP 3516
Legal HIV Menu Telephony EP     1603
```

Preprocessing Method – R (Meera)

- Forward-filled *activity name* and *ep name* to impute NA values
- Calculated the *ep name* journey of each call
 - Created variables representing **each preceding distinct ep name** per call
 - Last row of every call contains every ep it went through
- Sliced out the last row of every call
- Counted by *activity name*, *ep name*, and *all preceding ep names* to get the **count of each specific call journey**
- Created an *analysis* variable that extracts the most recent ep name within the specific menu options:
 - Farmworker, Benefits, Employment, Family, HIV, Housing, Immigration, Legal, Main, Other Legal, Pre-Legal Seniors, Intake Outdial*
- Counted by *analysis* and *activity name* → same results as Adam

contact_session_id	ep_name
4-6988-4f03-b5c1-5b533210a4ce	Main Number Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Main Number Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Main Number Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Main Number Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Main Number Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Main Number Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Pre-Legal Menu Seniors Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Pre-Legal Menu Seniors Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Legal Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Legal Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP
007c9ba4-6988-4f03-b5c1-5b533210a4ce	Closed Hours-Holidays Menu Telephony EP

last_distinct_before_last_ep	last_distinct_before_last_ep_prev	last_distinct_before_last_ep_prev2	last_distinct_before_last_ep_prev3
Closed Hours-Holidays Menu Telephony EP	Legal Menu Telephony EP	Pre-Legal Menu Seniors Menu Telephony EP	Main Number Telephony EP

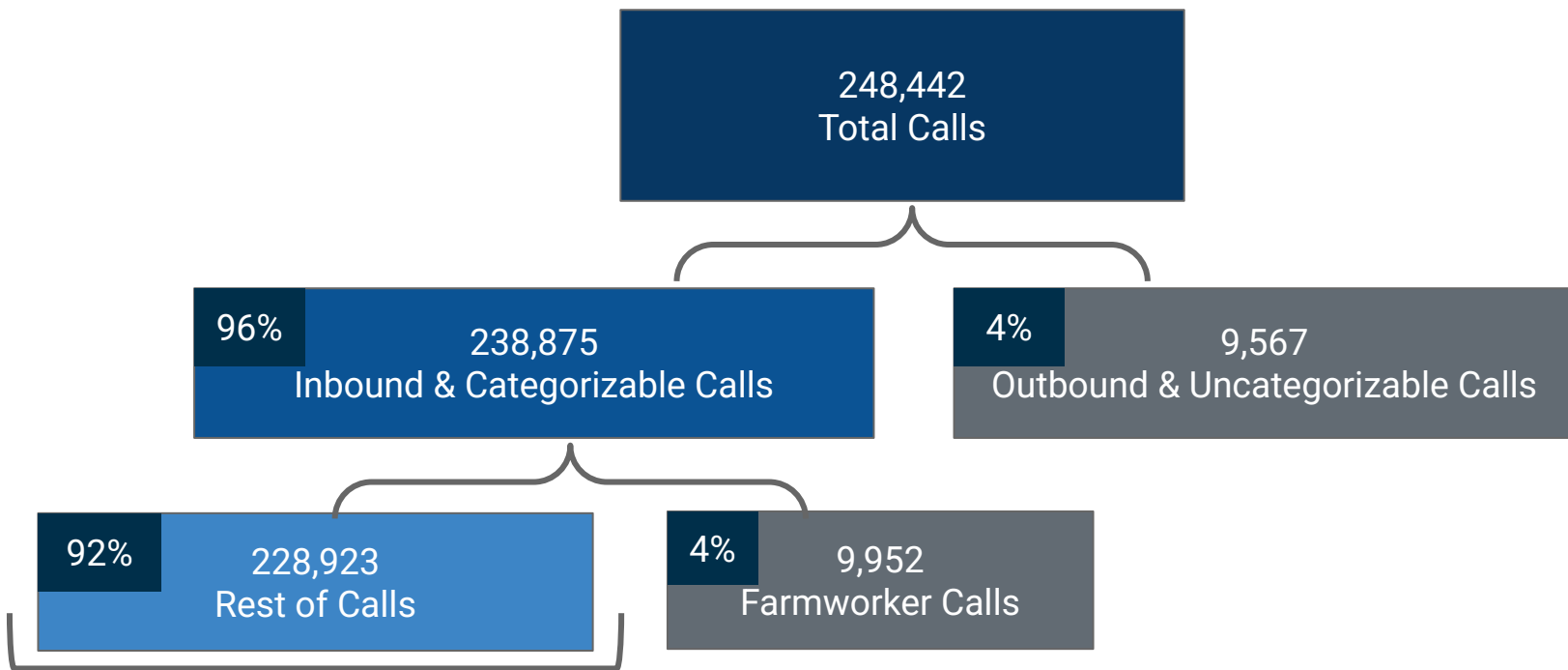
Outbound & Low Information Calls – Meera

1. Created a separate dataset for all calls with NA values for *activity name* throughout the whole call
2. Used the *termination reason* or *queue name* to categorize them through a “case when” statement
3. Counted by *ep name* and *activity name*
4. Added to spreadsheet to complete call categorization but excluded from abandonment calculations

ep_name	activity_name	n
Farmworker Main Number Telephony EP	No Activity	1
Intake Outdial EP	Intake Outdial Queue	2
Intake Outdial EP	System Error	3
Main Number Telephony EP	Customer Left	9
Main Number Telephony EP	NO_ANSWER_FROM_CUSTOMER	108
Main Number Telephony EP	No Activity	48
Main Number Telephony EP	System Error	1
Pre-Legal Menu Seniors Menu Telephony EP	NO_ANSWER_FROM_CUSTOMER	3
Pre-Legal Menu Seniors Menu Telephony EP	No Activity	2

No Activity Example	ep_name	flow_name	activity_name	timestamp	queue_name	agent_name	termination_reason
09be3dab-c9ed-4f0d-a123-b21942cd430e	Main Number Telephony EP	NA	NA	2024-12-16 13:18:55	NA	NA	NA
09be3dab-c9ed-4f0d-a123-b21942cd430e	Main Number Telephony EP	NA	NA	2024-12-16 13:18:55	NA	NA	NA

Total Analysis Figures



Abandonment Calculations

Spreadsheet Fill Process

From Code

Manual Fill

MENU	LAST ACTIVITY NAME	COUNT	Abandoned?	Reason	Person	Verified by (Optional)
Farmworker Main Number Telephony EP	FarmworkerMainMenu	9640	Abandoned	Left Menu	Adam	Meera
Farmworker Main Number Telephony EP	LanguageSelectionMenu	156	Abandoned	Left Menu	Adam	Meera
Farmworker Main Number Telephony EP	MigrantVoicemailTransfer	67	Not Abandoned	Voicemail Transfer	Adam	Meera
Farmworker Main Number Telephony EP	DisconnectContact1	44	Not Abandoned	Closed Queue / Menu	Adam	Meera
Farmworker Main Number Telephony EP	ClosedMenu	29	Not Abandoned	Closed Queue / Menu	Adam	Meera
Farmworker Main Number Telephony EP	ClinicVoicemailTransfer	7	Not Abandoned	Voicemail Transfer	Adam	Meera
Farmworker Main Number Telephony EP	StaffDirectorySpanishTransfer	5	Not Abandoned	Closed Queue / Menu	Adam	Meera
Farmworker Main Number Telephony EP	StaffDirectoryEnglishTransfer	3	Not Abandoned	Closed Queue / Menu	Adam	Meera
Farmworker Main Number Telephony EP	AddressFaxHoursMenu	1	Not Abandoned	Closed Queue / Menu	Adam	Meera
Farmworker Main Number Telephony EP	No Activity	1	Excluded	Undetermined	Meera	
Legal Benefits Menu Telephony EP	ClosedQueueMenu	4674	Not Abandoned	Closed Queue / Menu	Marcos	Adam
Legal Benefits Menu Telephony EP	ClinicVoicemailTransfer	818	Not Abandoned	Voicemail Transfer	Marcos	Adam
Legal Benefits Menu Telephony EP	LegalServerScreenPop	715	Not Abandoned	Callback (or Callback Attempt)	Marcos	Adam
Legal Benefits Menu Telephony EP	DisconnectContact1	334	Not Abandoned	Closed Queue / Menu	Marcos	Adam
Legal Benefits Menu Telephony EP	BenefitsMenu	294	Abandoned	Left Menu	Marcos	Adam
Legal Benefits Menu Telephony EP	VeteransBenefitsVoicemailTransfer	185	Not Abandoned	Voicemail Transfer	Marcos	Adam
Legal Benefits Menu Telephony EP	QueueMenu1	73	Abandoned	Left Queue	Marcos	Adam
Legal Benefits Menu Telephony EP	PlayMOH300s	41	Abandoned	Left Queue	Marcos	Adam
Legal Benefits Menu Telephony EP	ScreenPopProcessComplete	41	Not Abandoned	Callback (or Callback Attempt)	Marcos	Adam
Legal Benefits Menu Telephony EP	IntakePreQueueMessage1	17	Abandoned	Left Queue	Marcos	Adam
Legal Benefits Menu Telephony EP	DisconnectCallbackContact	8	Not Abandoned	Callback (or Callback Attempt)	Marcos	Adam
Legal Benefits Menu Telephony EP	CallbackRetry	3	Not Abandoned	Callback (or Callback Attempt)	Marcos	Adam
Legal Benefits Menu Telephony EP	LegalMenu2	3	Abandoned	Left Menu	Marcos	Adam
Legal Benefits Menu Telephony EP	ClosedMenu	1	Not Abandoned	Closed Queue / Menu	Marcos	Adam
Legal Benefits Menu Telephony EP	ConfirmCallbackNumber	1	Not Abandoned	Callback (or Callback Attempt)	Marcos	Adam
Legal Benefits Menu Telephony EP	PreQueueMessage2	1	Abandoned	Left Queue	Marcos	Adam
Legal Benefits Menu Telephony EP	StaffDirectoryEnglishTransfer	1	Not Abandoned	Closed Queue / Menu	Marcos	Adam
Legal Employment Menu Telephony EP	ClosedQueueMenu	3063	Not Abandoned	Closed Queue / Menu	Adam	Meera
Legal Employment Menu Telephony EP	ClinicVoicemailTransfer	536	Not Abandoned	Voicemail Transfer	Adam	Meera
Legal Employment Menu Telephony EP	LegalServerScreenPop	502	Not Abandoned	Callback (or Callback Attempt)	Adam	Meera
Legal Employment Menu Telephony EP	EmploymentMenu	182	Abandoned	Left Menu	Adam	Meera
Legal Employment Menu Telephony EP	DisconnectContact1	137	Not Abandoned	Closed Queue / Menu	Adam	Meera
Legal Employment Menu Telephony EP	QueueMenu1	33	Abandoned	Left Queue	Adam	Meera
Legal Employment Menu Telephony EP	ScreenPopProcessComplete	31	Not Abandoned	Callback (or Callback Attempt)	Adam	Meera

Common Reasons (to copy paste, for consistency)

Abandoned	Left Menu
Abandoned	Left Queue
Not Abandoned	Closed Queue / Menu
Not Abandoned	Callback (or Callback Attempt)
Not Abandoned	Voicemail Transfer
Not Abandoned	Call Ended
Not Abandoned	Not Served by Legal Aid
Not Abandoned	Transfer
Not Abandoned	Call Concluded
Not Abandoned	Received Info

Classifying Unclear Subgroups

Menu	Last Activity	Abandoned?	Reason	Person	Verified By
Legal Housing Menu Telpephony EP	StaffDirectoryEnglishTransfer	?	?	?	?



```
# 1. Find the Contact Session IDs that match the criteria
family_ids = df_final.loc[
    (df_final["Menu"] == "Legal Housing Menu Telephony EP") &
    (df_final["Last Activity Name"] == "StaffDirectoryEnglishTransfer"),
    "Contact Session ID"
].unique()

# 2. Pull ALL rows from df_main for these Contact Session IDs
test_output = df_main[df_main["Contact Session ID"].isin(family_ids)].copy()

test_output.head(200)
```



d44b132c-e870-4cd2-bc88-68ae441f6625	NaN	ClosedHoursHolidaysMenu	NaN	2024-05-02 16:34:17	NaN	NaN
d44b132c-e870-4cd2-bc88-68ae441f6625	Closed Hours-Holidays Menu Telephony EP	NaN	ClosedMenu	2024-05-02 16:34:17	NaN	NaN
d44b132c-e870-4cd2-bc88-68ae441f6625	Closed Hours-Holidays Menu Telephony EP	NaN	StaffDirectoryEnglishTransfer	2024-05-02 16:34:18	NaN	NaN
d44b132c-e870-4cd2-bc88-68ae441f6625	Closed Hours-Holidays Menu Telephony EP	NaN	NaN	2024-05-02 16:34:18	NaN	NaN
d44b132c-e870-4cd2-bc88-68ae441f6625	Closed Hours-Holidays Menu Telephony EP	NaN	NaN	2024-05-02 16:34:18	Staff Directory English Transfer	CBT Agent
d44b132c-e870-4cd2-bc88-68ae441f6625	Closed Hours-Holidays Menu Telephony EP	NaN	NaN	2024-05-02 16:35:03	Staff Directory English Transfer	CBT Agent
d44b132c-e870-4cd2-bc88-68ae441f6625	Closed Hours-Holidays Menu Telephony EP	NaN	NaN	2024-05-02 16:35:03	Staff Directory English Transfer	CBT Agent



Menu	Last Activity	Abandoned?	Reason	Person	Verified By
Legal Housing Menu Telpephony EP	StaffDirectoryEnglishTransfer	Not Abandoned	Call Concluded	Adam	Meera

Abandonment Results (Marcos / Sana)

Overall Call Abandonment Rate: 32.31%

Abandonment Rate by Menu, ordered by Abandonment Rate:

(could also maybe get rid of the other stuff and just have menu:
abandonment rate)

Menu Name	Total Calls	% of All Calls	Abandoned Calls	Abandonment Rate
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Benefits	7,210	3.15%	429	5.95%

Takeaways & Recommendations (Marcos / Sana)

Overall, high abandonment rates

Varies per menu

Even menus with small abandonment rates may not signify success on their own, lots of closed queues

Etc etc.

Works Cited

Python Code (Adam) (to be added):

R Code (Meera):

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Verbiage%20analysis/Meera%20Trivedi/Total_Abandonment_4Dec_Meera.R

Spreadsheet:

<https://docs.google.com/spreadsheets/d/1z8cZ-0Hxflcaw1d8fS9jQddN1noD1cVIIUNTUEO93uI/edit?gid=0#gid=0>